

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	4

Case Study Topic: Indian Uber Driver Dispatch with Negative-Edge Surge Incentives using Bellman-Ford Algorithm

Work Done Summary:

In this case study, the **Bellman-Ford Algorithm** was implemented to determine the shortest-cost route in an Uber driver dispatch network containing negative edge weights.

Unlike Dijkstra's Algorithm, Bellman-Ford can correctly handle negative-weight edges. In this scenario, negative weights represent **surge incentives (bonuses)** provided by Uber for drivers passing through high-demand pickup zones.

Implementation Details:

1. Directed Graph
2. Bellman-Ford Initialization
3. Iteration-wise Relaxation

Result:

Shortest Path :

IND → KOR → BTM → EC

Total Cost :

11

Output:

Shortest Cost to EC = 11

GitHub Link:

<https://github.com/saisharan134/DSA2-Case-Studies>

Student Signature	Faculty Signature